Transformer Model Tracking Algorithm

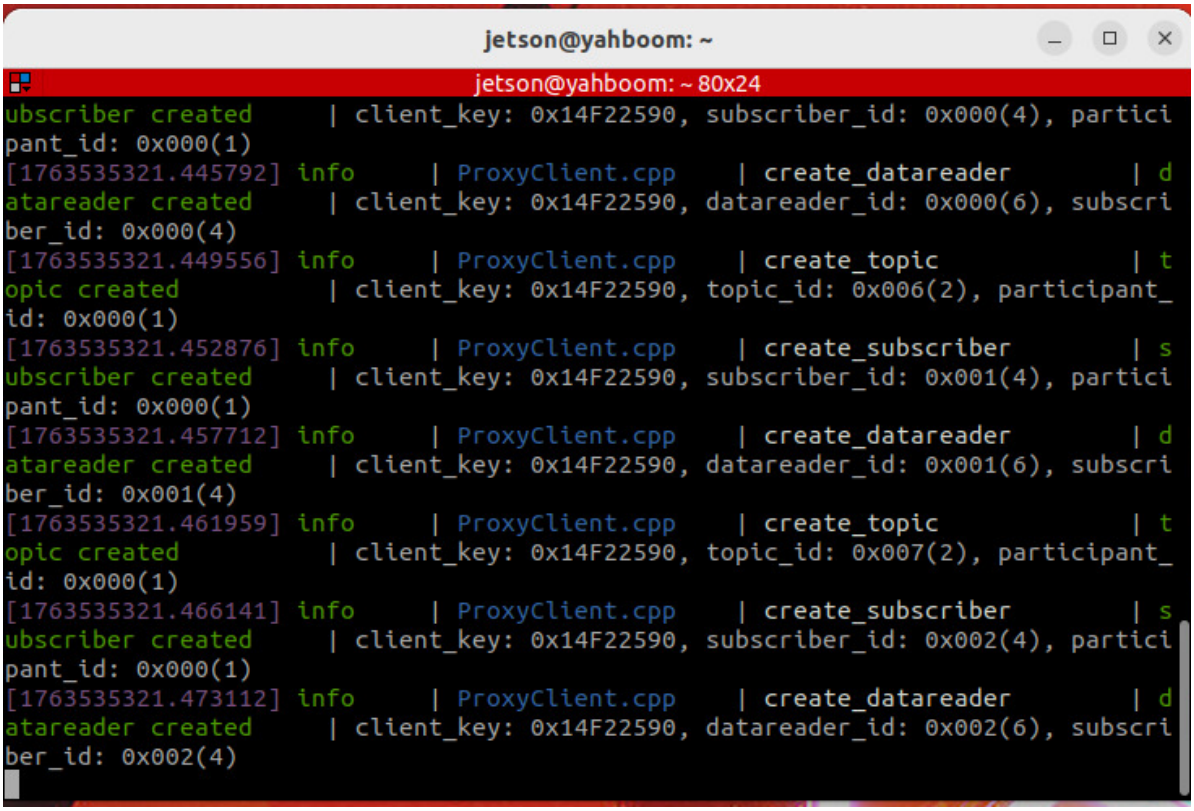
1. Course Content

- Learn about Transformer deep learning for target tracking

2. Preparation

- Start the microROS agent and connect to the chassis using the following command:

```
sh start_agent.sh
```



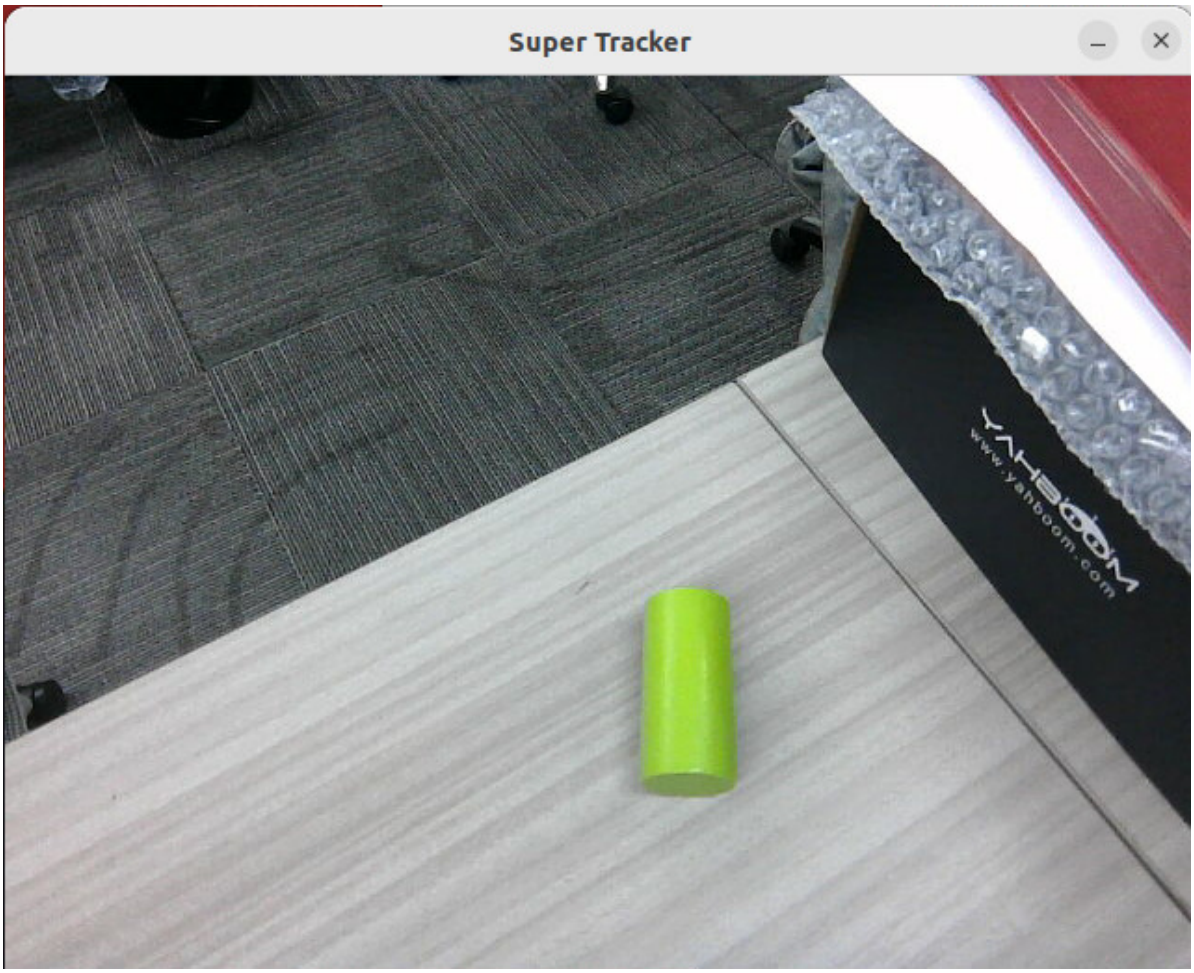
- Start basic nodes such as camera, robotic arm assistance, and radar fusion:

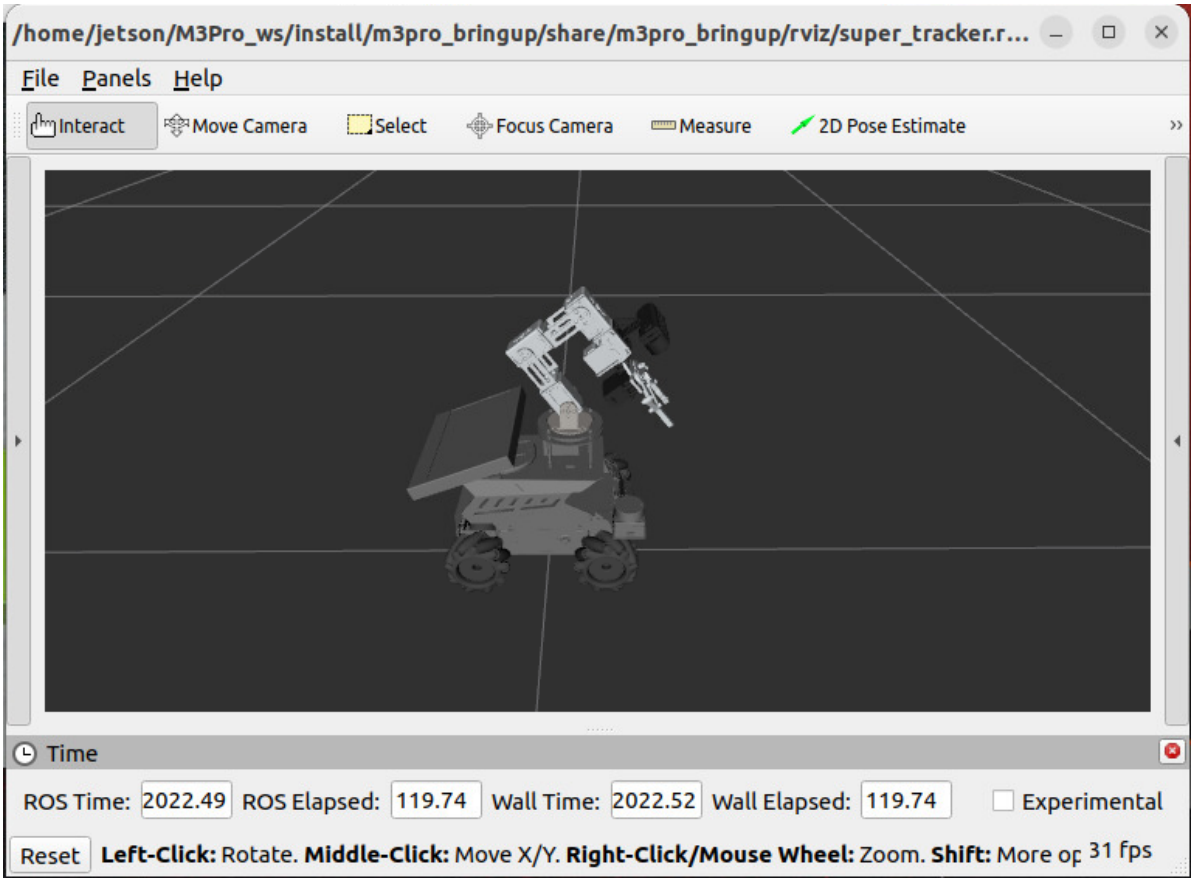
```
ros2 launch m3pro_bringup car_base.launch.py
```

3. Starting the Tracking Case

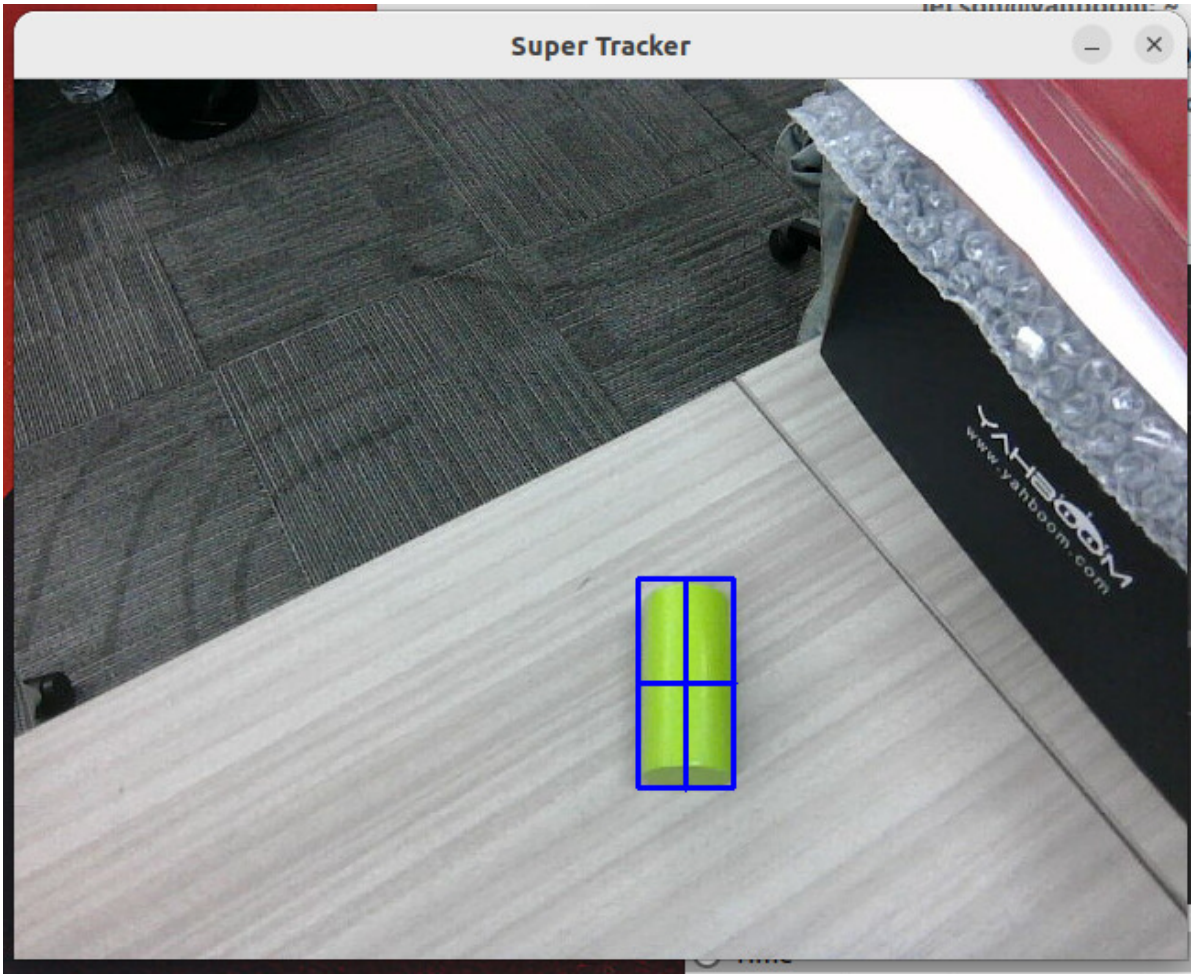
```
ros2 launch m3pro_bringup super_tracker.launch.py
```

- After the program starts, **Super Tracker window and rviz2 interface

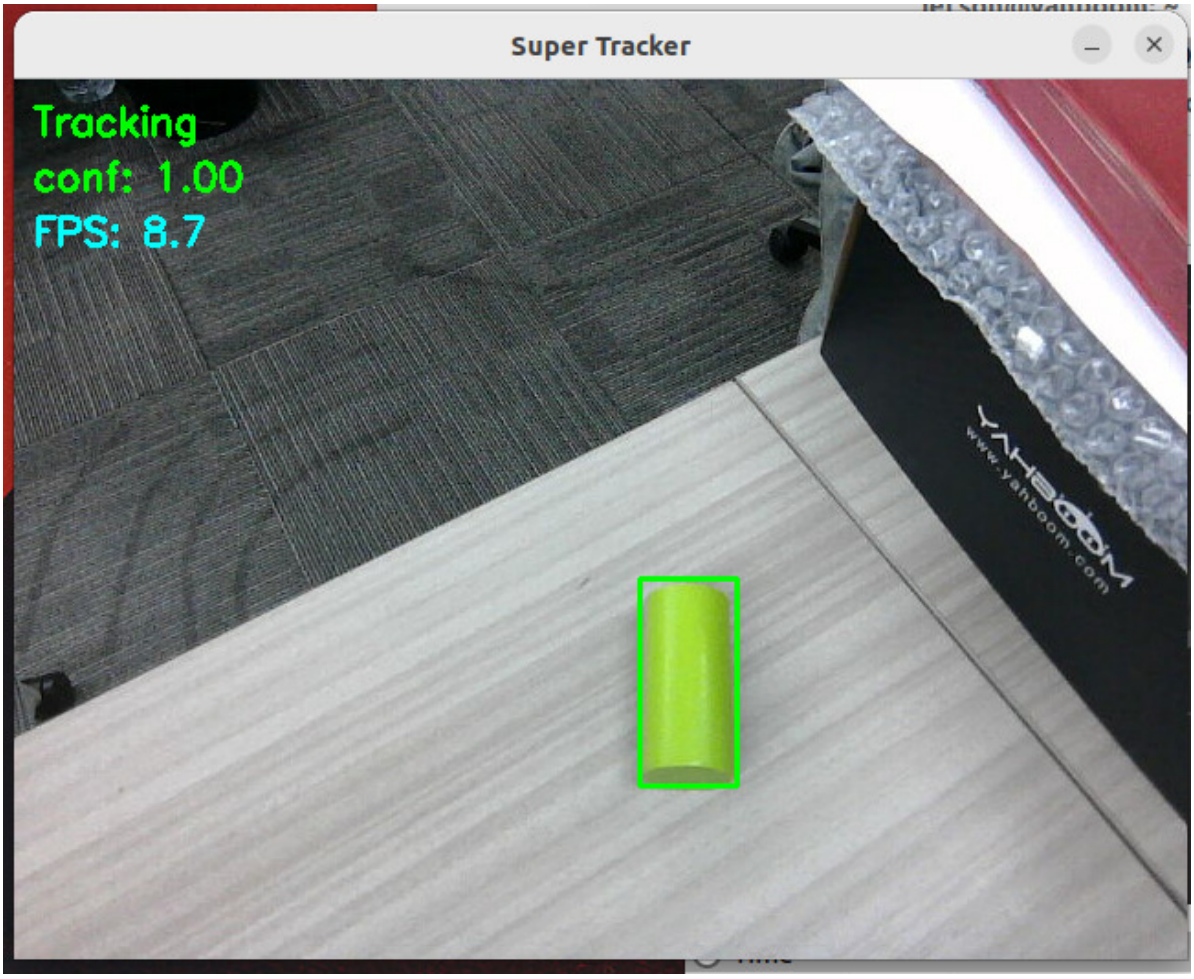


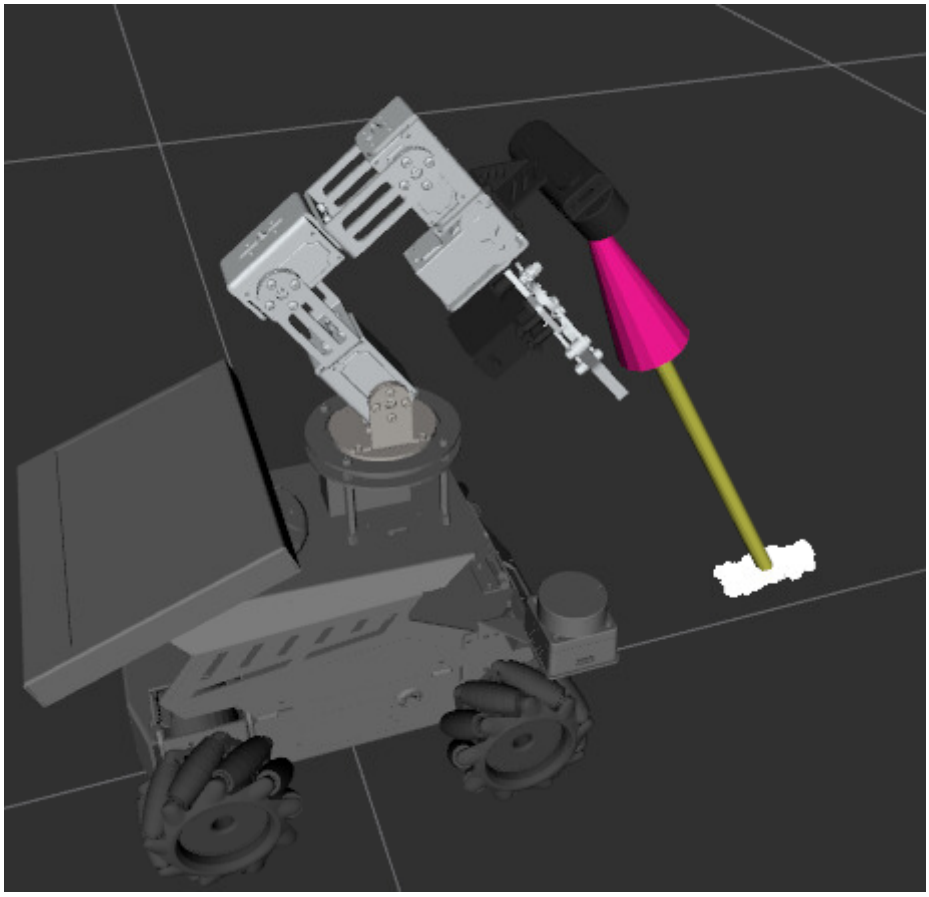


- In the Super Tracker window, press 'c' to enter selection mode. Use the mouse to select the target object you want to track, and press Enter to confirm.

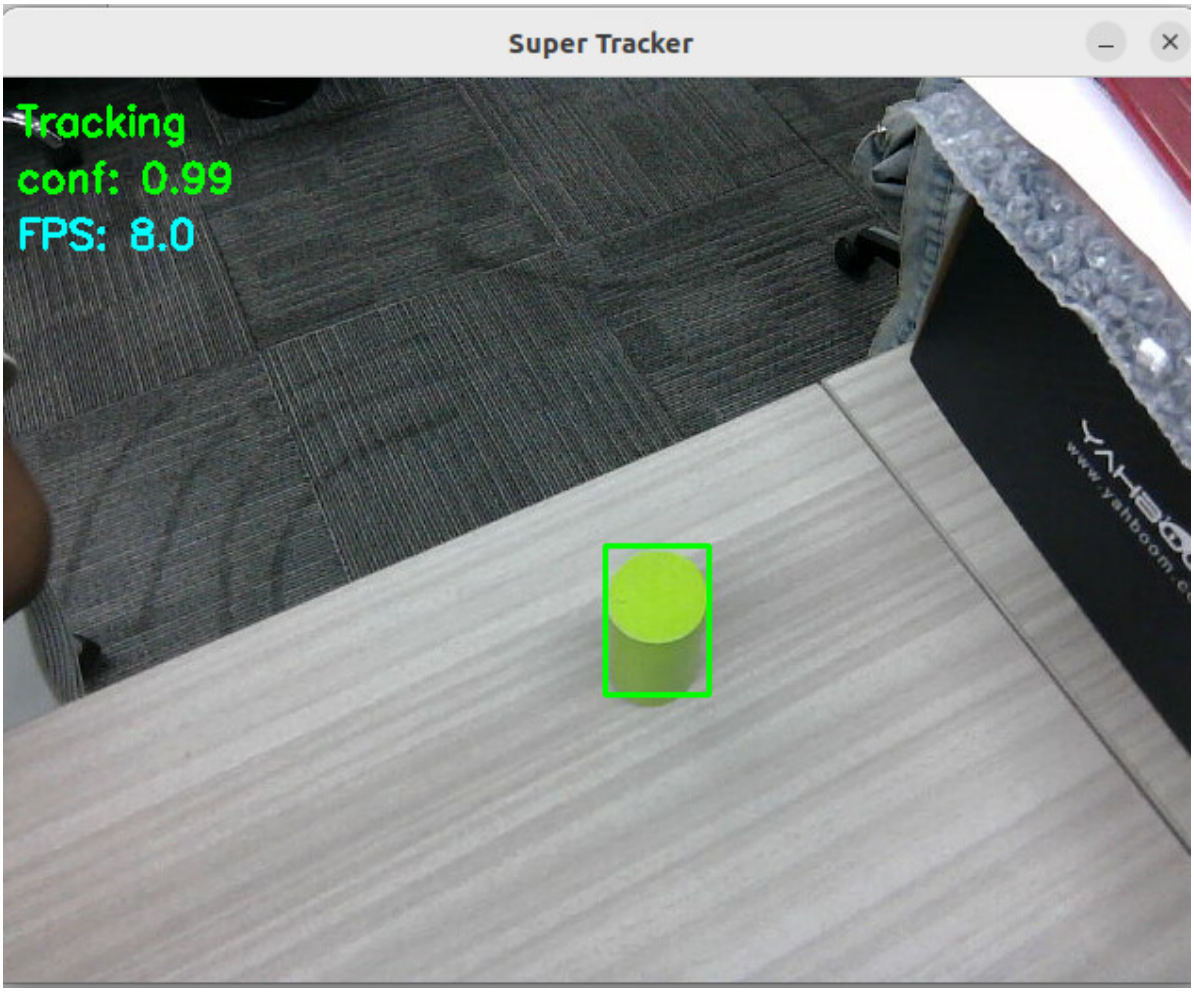


- Then it will enter tracking mode. In tracking mode, the point cloud of the target object will be segmented using PCL, which can be viewed in RViz.

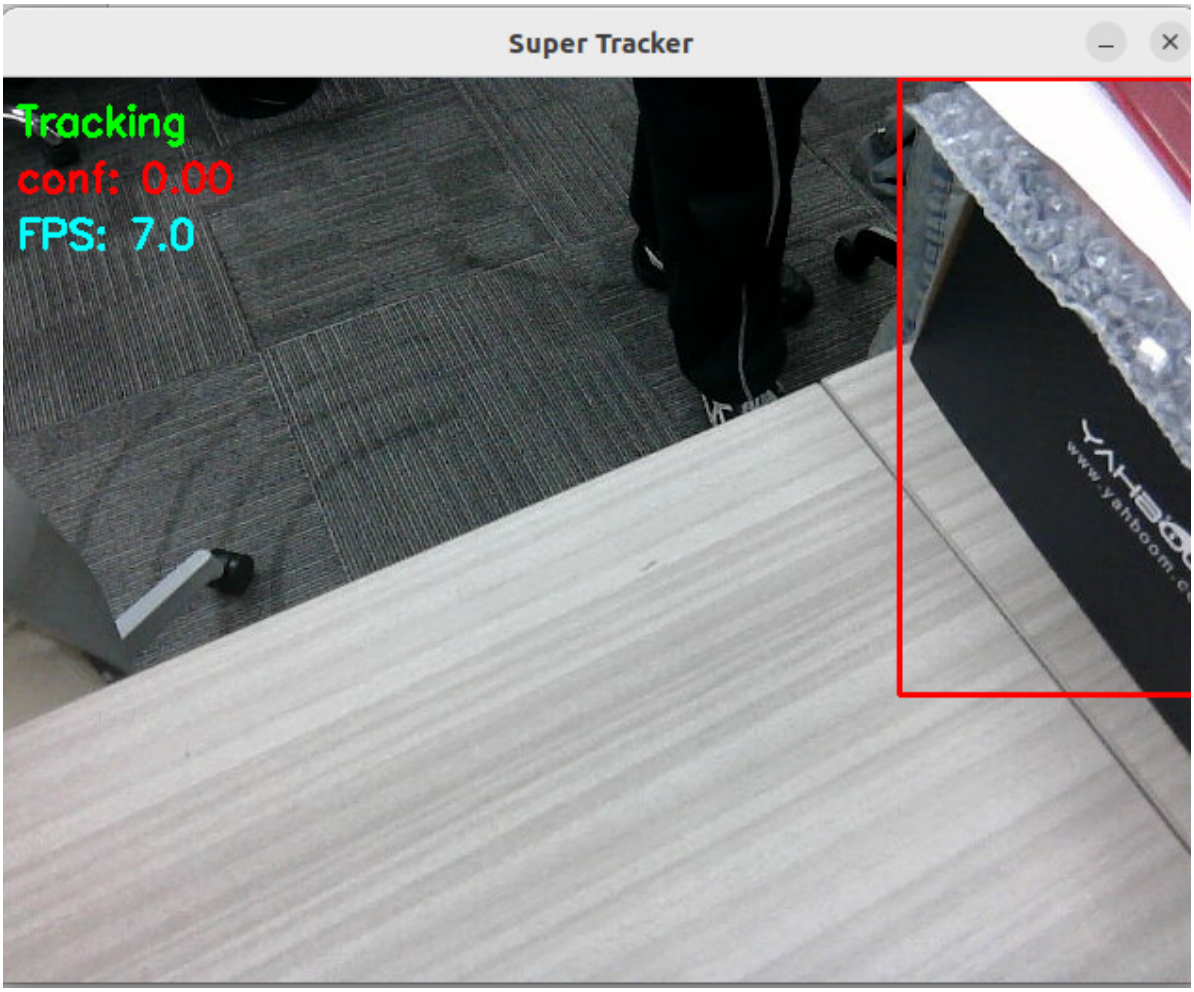




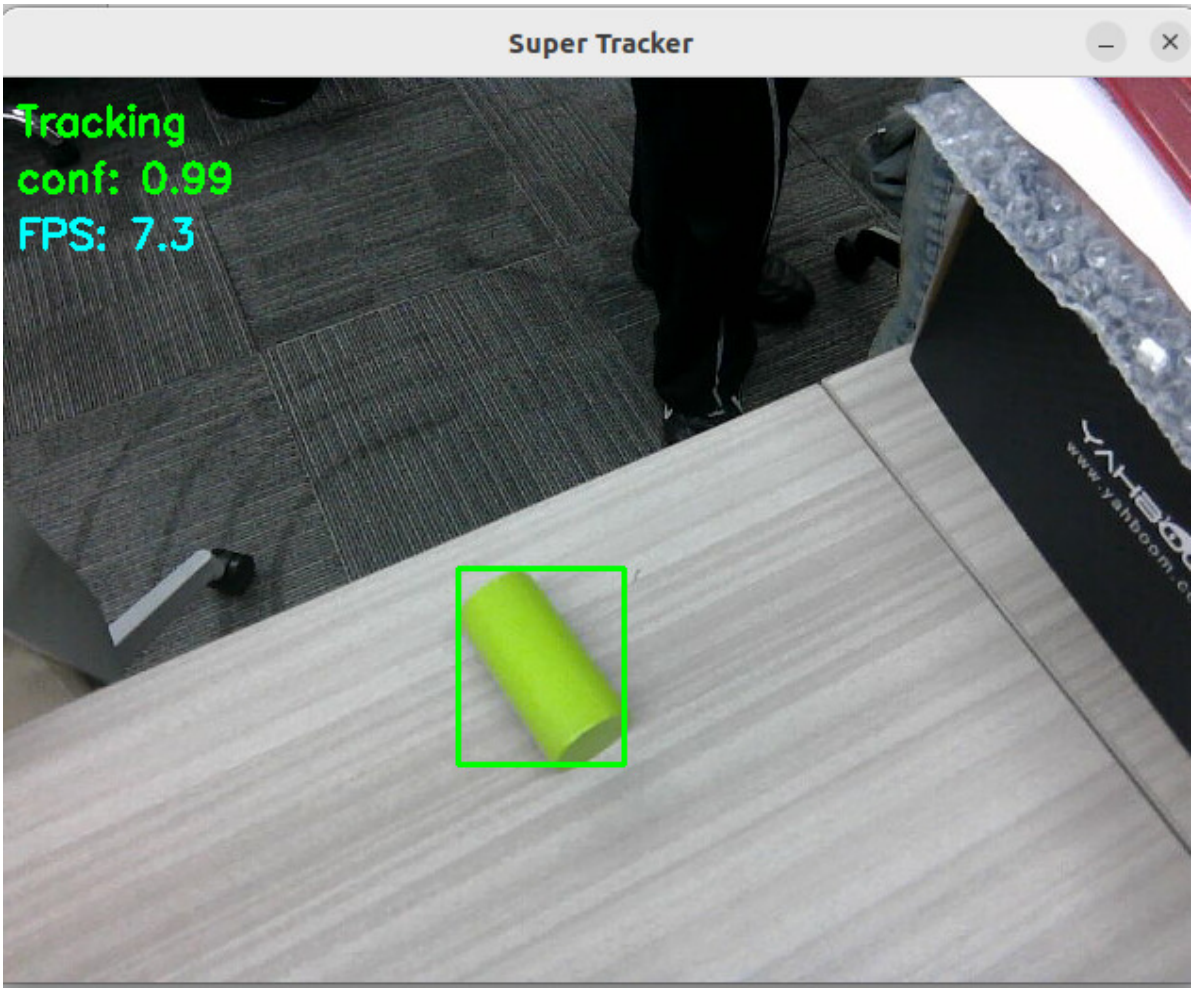
- When the camera's perspective relative to the tracked object changes, the tracking bounding box will adaptively deform to match.



- If the target's confidence score drops too low during tracking, the bounding box will turn red, and the system will begin searching for the target.



- When the target reappears within the field of view, the system will automatically reacquire it.



- During the tracking process, the node publishes the ROI bounding box coordinates and confidence score of the target object to the `/tracking_bbox` topic. You can view this data using the following command:

```
ros2 topic echo /tracking_bbox
```

```
timestamp:
  sec: 1778322787
  nanosec: 5830245
bbox:
- 228
- 269
- 316
- 374
confidence: 0.994836688041687
---
timestamp:
  sec: 1778322787
  nanosec: 165171703
bbox:
- 228
- 269
- 316
- 375
confidence: 0.996188759803772
```

- If you need to obtain the coordinates of the target object's point cloud centroid—specifically within the vehicle's `base_link` coordinate frame—you can query them using the following command:

```
ros2 run tf2_ros tf2_echo base_link roi_object
```

- In the output, "Translation" represents the x, y, and z coordinates of the target object's point cloud centroid within the `base_link` coordinate frame.

```
jetson@yahboom:~$ ros2 run tf2_ros tf2_echo base_link roi_object
[INFO] [1778323060.476885179] [tf2_echo]: Waiting for transform base_link ->
_link" passed to canTransform argument target_frame - frame does not exist
At time 1778323061.382172971
- Translation: [0.262, 0.020, 0.061]
- Rotation: in Quaternion [-0.662, 0.675, -0.229, 0.231]
- Rotation: in RPY (radian) [-2.479, 0.010, -1.586]
- Rotation: in RPY (degree) [-142.029, 0.553, -90.895]
- Matrix:
 -0.016 -0.788  0.615  0.262
 -1.000  0.018 -0.002  0.020
 -0.010 -0.615 -0.788  0.061
  0.000  0.000  0.000  1.000
```

- When you need to stop tracking an object, you can do so by making a service request:

```
ros2 service call /cancel_track smart_tracker/srv/CancelTrack "{}"
```

```
jetson@yahboom:~$ ros2 service call /cancel_track smart_tracker/srv/CancelTrack "{}"
requester: making request: smart_tracker.srv.CancelTrack_Request()

response:
smart_tracker.srv.CancelTrack_Response(success=True, message='Tracking canceled')
```

> [!IMPORTANT]
>
> - In the **[OpenClaw Embodied Intelligence in Action]** chapter, the deep learning-based tracking module discussed here will be utilized for robotic arm target tracking and object grasping tasks.

4. Source Code Analysis

- Path to the `super_tracker` launch file:

```
$HOME/M3Pro_ws/src/m3pro_bringup/launch/super_tracker.launch.py
```

```
import os
from launch import LaunchDescription
from launch.actions import DeclareLaunchArgument
from launch.substitutions import LaunchConfiguration
from launch_ros.actions import Node
from ament_index_python.packages import get_package_share_directory
def generate_launch_description():

commom_param_file=os.path.join(get_package_share_directory('m3pro_bringup'),'con
fig','common.yaml')
super_track_model=os.path.join(os.path.expanduser('~'),'MODELS','tracker','super
_track_sim.onnx')
rviz_config=os.path.join(get_package_share_directory('m3pro_bringup'),'rviz','su
per_tracker.rviz')
declared_arguments = []

declared_arguments.append(
DeclareLaunchArgument(
"display_windows_only_track",
default_value='false',
description="",
)
)
declared_arguments.append(
DeclareLaunchArgument(
"display_windows",
default_value='true',
description="",
)
)
```

```
# tracker node
tracker_node = Node(
package='smart_tracker',
executable='super_tracker',
name='super_tracker',
parameters=[{
'display_windows_only_track': LaunchConfiguration('display_windows_only_track'),
'display_windows': LaunchConfiguration('display_windows'),
'model_path': super_track_model,
}],
remappings=[('/image_raw', '/camera/color/image_raw') ],
output='screen'
# prefix='gdb -ex run --args'
)

# PCL segmentation node for ROI point cloud segmentation
pclsegment_node = Node(
package='smart_tracker',
executable='pcl_segment',
name='pcl_segment',
parameters=[commom_param_file],
output='screen'
)
rviz_node = Node(
package = 'rviz2',
executable = 'rviz2',
name='rviz2',
arguments=['-d', rviz_config],
output='screen',
)
# Create launch description
return LaunchDescription([
*declared_arguments,
tracker_node,
pclsegment_node,
rviz_node
])
```

5. Attachments

- The .deb package required for the `super_tracker` functionality (**Annex — .deb Functional Package**) and the ONNX model file (**Annex — Model File**) can be found in the tutorial attachments.
- Functional Package: `ros-humble-smart-tracker_0.0.0-0jammy_arm64.deb` (Install using `apt install`)
- Model File: `super_track.onnx`